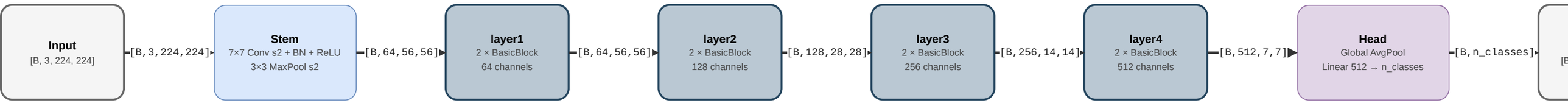
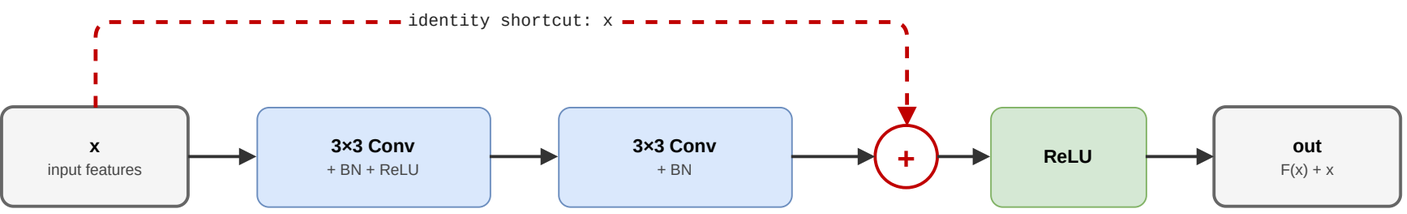
ResNet-18 — residual learning · input 224×224, output n_classes logits

Input / Output | Conv (+ BN, activation) | Group of blocks | Pool + classifier | **red dashed line = skip connection**

1. Overall architecture



2. Inside one BasicBlock (the repeating unit)



Key points for viva

- The block learns a residual F(x) and outputs F(x) + x, so gradients flow through the skip path and deep nets train easily.
- 4 stages (64 → 128 → 256 → 512 channels); each stage after the first halves the resolution (stride 2) and uses a 1×1 conv on the skip path to match shapes.
- 8 BasicBlocks × 2 convs + stem conv + FC = 18 weight layers · ≈ 11.2 M parameters.